Views

1. What is a View?

A view is a virtual database object that contains a SQL query, and does not physically store data. When users query the View, Oracle gets data from the underlying tables using the View definition. Views make data easier to access and provide a personalized view of information.

2. Why would companies use Views?

Views are used by companies to simplify complex queries, provide consistent data access, and hide unnecessary table details from users. Views simplify queries and give different departments access to only the information relevant to their work.

3. What are the security benefits of Views?

Views improve security by limiting access to sensitive data. Users can be given access to a View, but not directly to the underlying tables. This allows organizations to hide confidential columns such as salary, personal information or financial data.

4. Difference between Table, View and Materialized view Table

A Table is a physical database object that contains actual data rows and columns. The database stores data permanently. Data can be inserted, updated and deleted.

A View is a virtual object specified by a query. It does not store data itself and always shows the latest data from the underlying tables.

Materialized View stores the result of the query physically . It boosts performance for complex queries because data is precomputed and refreshed periodically instead of being calculated on the fly.

5. Perspectives: three enterprise use cases

Human resource staff can see employee info, but salary info is hidden.
Easy dashboards for managers using data from different tables.
Enable reporting teams to query business data via a controlled interface, rather than directly to the base tables.

Stored Procedures

1. What is Stored Procedure?

A Stored Procedure is a named block of SQL and PL/SQL code stored in the database and executed when needed. It can take input parameters, perform operations and return results.

2. Why do businesses use Procedures?

Procedures allow companies to encode business logic in one place, the database. This eliminates duplicate code, improves consistency and makes maintenance easier as changes will only have to be made in one place.

3. What do procedures solve?

Procedures cover a variety of business problems including:

Duplicated SQL code in applications.
Inconsistent business rules
Complex queries are hard to maintain.
Reduced application performance for sending multiple SQL statements.

4. Procedure vs Function Procedure

A Procedure is a task or business operation. It may take parameters . It may return values through output parameters . But its main purpose is to perform actions .

A Function has to return a value. It is used widely in SQL statements and calculations where a result is required.

5. Enterprise procedures: three use cases

How to Calculate Yearly Employee Bonuses

Processing monthly payroll transactions.

Business rules driven employee performance reports generation.

Triggers

1. What is trigger?

A Trigger is a database object that automatically executes when a specific event occurs, such as an INSERT, UPDATE, or DELETE on a table.

2. What is the purpose of Triggers for companies?

Triggers are used by companies to enforce business rules, track audit records and act immediately on changes made to the database without any manual intervention.

3. Difference between BEFORE Trigger and AFTER Trigger

BEFORE Trigger
Runs before the database operation is run. It is often used for validation and for business rules validation prior to the data being saved.

AFTER Trigger
Executes after the database operation is successfully done. It is usually used for auditing, logging and notifications.

4. What are the risks of overuse of Trigger?

Lower database performance.
More complexity to the system.
Difficult to debug and maintain.
Multiple triggers can interact in unexpected chain reactions.

5. Three Triggers use cases for enterprise

Audit table for recording salary changes.
Automated user activity logging.
Alerts for critical data changes.

Scheduler Jobs

1. What is a Scheduled Job?

A Scheduler Job is a database task that runs automatically at a specific date and time or on a regular interval. It helps automate routine database activities and removes the need to perform them manually.

2. Why do companies use Scheduler Jobs?

Companies use Scheduler Jobs to automate repetitive tasks, increasing efficiency, reducing human error and ensuring that critical processes are run on time.

3. Difference between Trigger and Scheduler Job Trigger

A Trigger is automatically executed when a specific database event occurs, such as an INSERT, UPDATE, or DELETE operation.

Scheduler Job executes according to a preset schedule, say every day, every week or every month, irrespective of user activity.

4. What processes are commonly automated?

- Report generation.
- Database backups.
- Data cleanup and archiving.
- Sending notifications and alerts.
- Batch processing tasks.

5. Three enterprise use cases for Scheduler Jobs

1. Generating weekly management reports.
2. Running nightly database backup processes.
3. Sending monthly employee performance summaries.

Scenario 1

Object: View

A View can display only Employee Name and Department Name while hiding salary information. This provides controlled access and improves security.

Scenario 2

Object: Trigger

A Trigger can automatically record every salary update into an audit table whenever a salary change occurs.

Scenario 3

Object: Scheduler Job

The report must be generated automatically every Friday at 4:00 PM, which is a time-based task best handled by a Scheduler Job.

Scenario 4

Object: Procedure

A Procedure can store the bonus calculation logic in one reusable database object and be executed whenever required.

Scenario 5

Object: Trigger

A Trigger can immediately detect salary modifications and initiate a notification process automatically.

Scenario 6

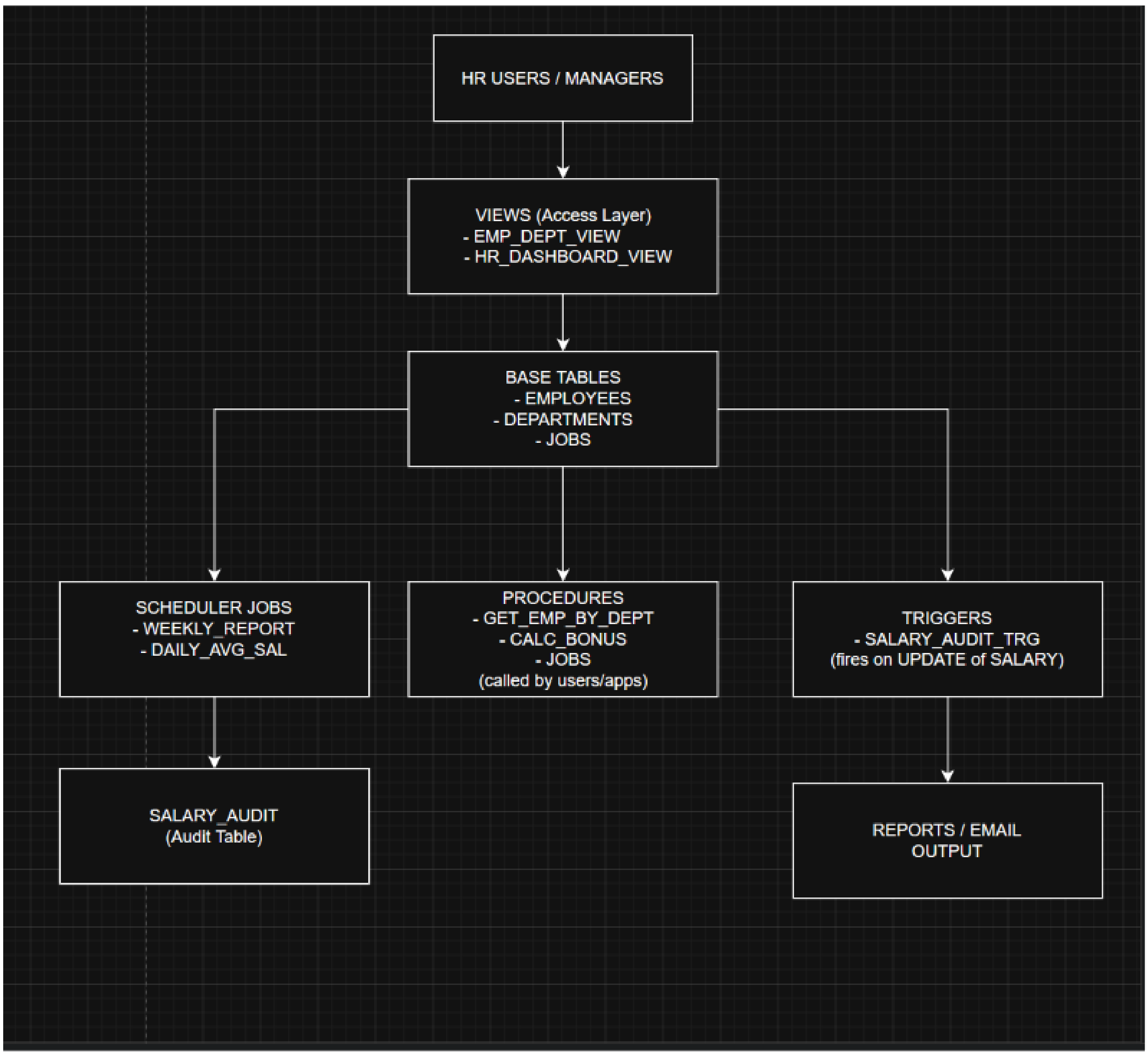
Object: View

A View provides controlled access to employee information while hiding the underlying tables and protecting sensitive data.

Description of Architecture

The HR system makes use of Views, Procedures, Triggers, and Scheduler Jobs to ensure secure access and automation of business process and maintenance of audit trails.

- HR Users view employee details using Views rather than directly querying the tables.
- Procedures execute the business processes like computation of bonuses and generating employee reports.
- Triggers automatically log any salary increments in the audit tables.
- Scheduler Jobs create reports based on specified schedule times.
- Employees are logged in HR tables, and salary details are logged in audit tables.



Reflection

Question

If applications can perform all business logic themselves, why do enterprise systems still place logic inside the database?

Answer

The reason behind why enterprise systems still implement critical business logic within the database is that it enhances security, consistency, efficiency, and maintainability.

The use of views enables the provision of access to data while safeguarding sensitive tables. Using procedures helps centralize the implementation of business logic, which allows the consistent execution of logic within all applications. The implementation of triggers ensures auditing and monitoring of activities whenever changes occur in the database. The use of Scheduler jobs automates regular business functions like generating reports and performing other maintenance activities.

This way, business logic can be centralized within the database, and thus there will be no differences in applying business rules by all the applications accessing the data.